



'all off' button. Each individual LED can also be made to glow/turn off by ticking the corresponding checkbox. The LED control application program works without external power supply in the circuit.

There is another application program used to control the number

of rotations, direction (clockwise or anticlockwise) and rotational speed of the stepper motor. Optocoupler MCT2E ICs (IC2 through IC5) are recommended for interfacing the stepper motor to the parallel port. Each of these ICs has an internal LED and a photo-transistor.

The external circuitries of the internal LED and photo-transistor of IC MC12E differ. The internal LED is driven by the signal from the parallel port, whereas a separate 12V supply powers the photo-transistor (refer Fig. 4). Therefore it completely isolates the parallel port from the rest of the interfacing circuit. This ensures safety of the parallel port as it can get damaged in case of shorting of the external circuit.

IC ULN2003 (IC6) is used to drive the stepper motor. It is a high-voltage, high-current Darlington transistor array. It consists of seven npn Darlington pairs that feature high-voltage outputs with internal common-cathode clamp diodes for switching inductive loads. The collector-current rating of a single Darlington pair is 500 mA. The Darlington pairs can be paralleled for higher current capability. Other applications of this IC include relay driver, lamp driver, display driver, line driver and logic buffer.

Stepper motor

Different types of stepper motors with different specifications are available in the market. The one used here is a